

Loan Approval Outcome Prediction

Data-Driven Strategies for Smarter Lending Decisions



Presented by:
Maryam Isa
Shaikha Ali Abdali
Zahraa Mohammed

The Challenge of Loan Decisions



The Problem

Loan defaults and rejections cost financial institutions billions annually, with inaccurate lending decisions affecting both profitability and access to credit for reliable borrowers. Using demographic, financial, employment, and credit history data from **47,000+** loan applicants, can we build a **supervised binary classification** model to predict whether a loan will be **approved or rejected** — and identify which factors drive that outcome most?

Because approvals make up only 22% of the dataset, accuracy alone is misleading. We optimise for F1-Score, balancing Precision (avoiding wrong approvals) and Recall (not missing genuine ones).

What We're Working With

47,130

Loan Applicants

after cleaning

14

Features per Applicant

age, income, credit score, and more

78% Rejected

22% Approved

Class imbalance — F1 is our metric

Features Used

Age · Gender · Education Level · Annual Income · Employment Experience
Home Ownership · Loan Amount · Loan Intent · Interest Rate
Loan % of Income · Credit History Length · Credit Score · Prior Loan Defaults

★ Prior Loan Defaults

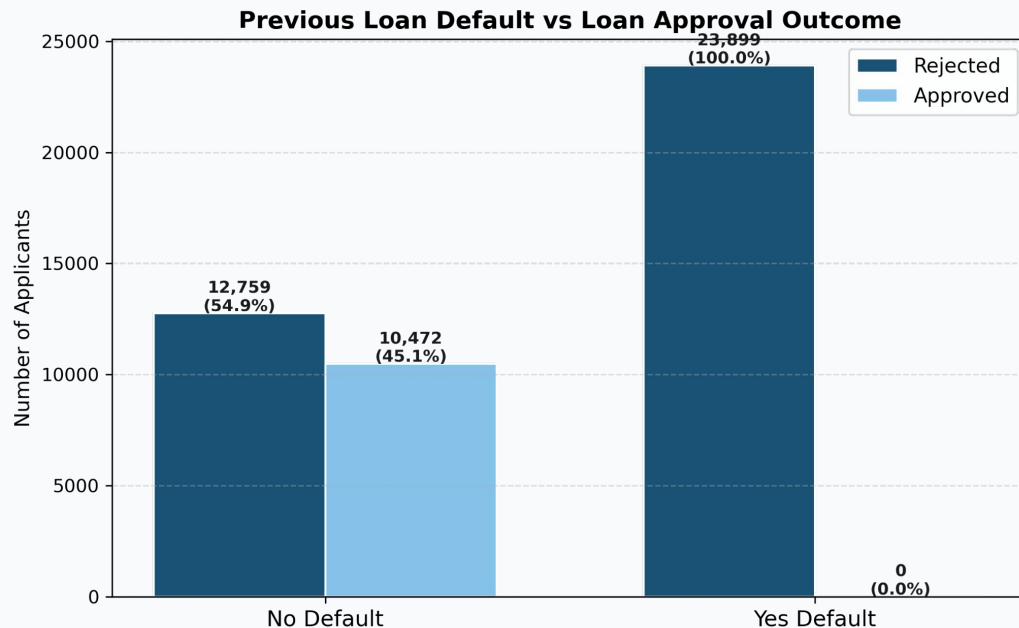
Turned out to be the single strongest predictor in the entire dataset — **every approved applicant had a clean record.**

What the Data Told Us



Prior Defaults = Automatic Rejection

Not a single approved applicant had a prior default on file. **All 23,953 applicants with a default history were rejected** — the single strongest predictor in the dataset.



What the Data Told Us



Income Alone Doesn't Decide

Higher earners were less likely to get approved (correlation: -0.17). Wealthier people tend to apply for larger, riskier loans — so **income without context is misleading.**



No Single Feature Dominates

Credit score had almost zero linear correlation with approval (-0.01). The **model needs to combine multiple signals to predict well.**
eg. interest rate, loan-to-income.

Cleaning & Feature Engineering



Removed Impossible Values

Dropped ages **over 70** and work experience **over 60**. Only few rows affected — less than 0.04% of the data.



Fixed Income & Loan Columns

Stripped \$ signs, **converted to numbers**. Also **removed 104 income outliers** above \$500,000.



Standardised Categories

Fixed mixed casing and extra spaces — 'MALE', 'male', ' Male ' all became **one consistent value**.



Encoding Categorical Variables

Gender & prior defaults → binary (0/1)
Education → ordinal (High School = 0 to Doctorate = 4)
Home ownership & loan intent → one-hot encoding (no natural order)



Prevented Data Leakage

Scaler fitted on training data only, applied to test data — the model never saw what it shouldn't.

Two Models, One Goal

Model A



Logistic Regression

Correctly approved **1,542** applicants and correctly rejected **6,901**. Missed **552** real approvals and wrongly approved **431** people who shouldn't have been.

- **Accuracy: 89.6%** — but inflated by the class imbalance
- **Precision: 78.2%** — when it said approved, it was right 78% of the time
- **Recall: 73.6%** — caught 73.6% of actual approvals
- **F1-Score: 75.8%**

Model B



K-Nearest Neighbors

Correctly approved **1,478** applicants and correctly rejected **6,990**. Missed **616** real approvals but had fewer wrong approvals (**342** vs 431 in LR).

- **Accuracy: 89.8%** — similar to LR overall
- **Precision: 81.2%** — more careful about who it approves
- **Recall: 70.6%** — but missed more genuine approvals than LR
- **F1-Score: 75.5%** — slightly lower than LR despite better precision

Performance Metrics – All 4 Models

Accuracy

Out of everyone, how many did we get right?

Precision

When we said Approved, how often were we right?

Recall

Out of real approvals, how many did we catch?

F1-Score

Balance between Precision and Recall.

Model	Accuracy	Precision	Recall	Specificity	F1-Score
Baseline Logistic Regression	89.57%	78.16%	73.64%	94.12%	75.83%
Tuned Logistic Regression	89.61%	78.17%	73.88%	94.11%	75.96%
Tuned 2 Logistic Regression	85.60%	62.01%	90.83%	84.11%	73.71%
Baseline KNN (k=5)	89.84%	81.21%	70.58%	95.34%	75.52%
★ Tuned KNN	91.11%	85.89%	71.78%	96.63%	78.20%

How We Made the Models Better

What We Did



GridSearchCV

We gave the computer **hundreds of setting combinations** to test automatically. It ran each one and **picked the highest-scoring setup**.



Pipeline

We chained the scaler and model together so data prep happens inside each test — **keeping evaluation fair and preventing leakage**.



Regularisation

Applied to Logistic Regression to stop it relying too heavily on any one feature. **Best setting: C=1 with L2 penalty**.

What Changed

Tuned KNN ★ **Biggest Improvement**

Best settings: **k=15, Manhattan distance, distance weighting**

F1: 75.52% → **78.20%** Precision: 81.21%
→ **85.89%**

120 fewer false predictions vs baseline KNN

What This Means for the Business



Best Model

Tuned KNN - 91.1% Accuracy, 85.8% Precision

Improved classification performance by **reducing False Positive and False Negative predictions by 120 instances.**



Key Finding

Prior Defaults Drive Everything

Every approved applicant had a clean default history. Flag applicants with defaults for manual review — **don't auto-reject, but do investigate further.**



Business Actions

Three Actionable Steps

- ① Use **F1-Score** and **not accuracy** to measure lending models.
- ② Treat prior defaults as a red flag, not automatic rejection.
- ③ Use loan_income ratio as a standard underwriting rule.

**Thank you
for listening!**